Final Take Home Exam

Adam Kaderbhai

Problem 1

a.)

`Variety` is a categorical variable since it does not involve numerical measurements And it has a Nominal level of measurement since it does not have a specific order to it

b.)

```r
quantile(onions$Dens)
```

```
     0%     25%     50%     75%    100% 
 2.1400  4.6475  8.6950 18.4825 31.0800 
```

```r
min <- 2.14
Q_1 <- 4.6475
Median <- 8.6950
Q_3 <- 18.4825
max <- 31.0800


IQR <- Q_3 - Q_1
IQR
```

```
[1] 13.835
```

- min $= 2.14$
- $Q_1 = 4.6475$

- Median = 8.6950
- Q_3 = 18.4825
- max = 31.08
- IQR = 13.835

c.)

```r
quantile(onions$Yield, .60)
```

```
  60%
52.92
```

60% of the onion yields are less than or equal to 52.92 units.

d.)

```r
point_estimate <- mean(onions$Yield)
point_estimate
```

```
[1] 54.71
```

```r
standard_dev <- sd(onions$Yield)

n <- 30

C_level <- qt(.975, df = 29)

Margin_of_Error <- C_level * (standard_dev/sqrt(n))
Margin_of_Error
```

```
[1] 12.4335
```

```r
point_estimate - Margin_of_Error
```

```
[1] 42.2765
```

```r
point_estimate + Margin_of_Error
```

```
[1] 67.1435
```

This is a t-distribution since the population standard deviation is unknown. The point estimate which is also the sample mean is 54.71. The Margin of error is 12.4335. And the endpoints are 42.2765 and 67.1435

Problem 2

a.)

In order to find the coefficient of correlation, the variables need to be quantitative and need to have a linear relationship. Variables that have a linear relationship but zero correlation look like random clouds.

b.)

```r
cor(cheese_sample$Lactic, cheese_sample$Taste)
```

```
[1] 0.5133197
```

This shows a positive moderate correlation between the variables `Lactic` and `Taste`.

c.)

```r
lm(cheese_sample$Taste ~ cheese_sample$Lactic)
```

```

Call:
lm(formula = cheese_sample$Taste ~ cheese_sample$Lactic)

Coefficients:
         (Intercept)  cheese_sample$Lactic
              -2.644                15.921
```

$cheese\_sample\$Taste = -2.644 + 15.921 cheese\_sample\$Lactic$

For every 1 unit increase of Lactic , we expect the taste to increase by approximately 15.92 units, on average.

d.)

This calculation is appropriate because the lactic acid concentration of `1.30` falls within the range of all lactic acid concentrations

```
Taste_predicted = -2.644 + (15.921 * 1.30)
Taste_predicted
```

```
[1] 18.0533
```

The predicted taste rating for the cheese with lactic acid concentration of 1.30 is 18.0533

e.)

```
Taste_actual <- 25.9
Taste_predicted <- 18.0533

Residual = Taste_actual - Taste_predicted

Residual
```

```
[1] 7.8467
```

The cheese with lactic acid concentration of 1.30 has a taste rating of 7.8467 units greater then what we would expect for the cheese with that lactic acid concentration.